Case Description – recruitment process

Hiring a new associate or full professor at NTNU, is a process that involves several steps and actors. Before we describe this process, it is important to remember the organizational structure of the university. The university consists of 8 faculties and each faculty has a number of institutes. Each institute consists of a number of groups.

The hiring process is initiated when an institute determines that it wants to hire a new staff member. The reason for this could be that someone left or that a new position is created for strategic reasons. Based on the required profile, the institute writes an advertisement text for the position. This text is sent to the faculty, where it is assessed by the recruitment committee. This committee decides whether the position can be announced by checking it against the budget and the faculties recruitment plan for the coming five years.

If the position can be announced, it is put on Jobbnorge.no. Applicants can use this portal to apply for the advertised position. They upload all the required documents, such as cv, education certificates and scientific publications they want to be included in the assessment. After the deadline has passed, the application is closed.

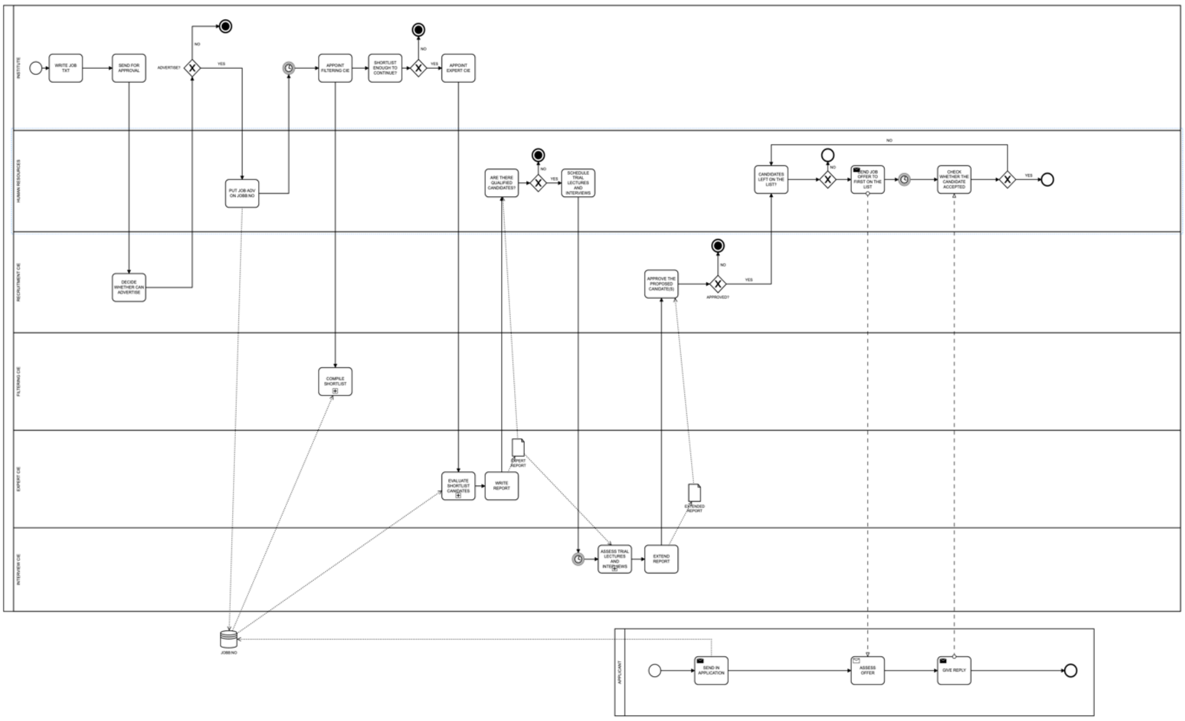
The institute now appoints a so-called ‘filtering’ committee that consists of two people. They go through the list of applicants, and remove those that don’t have a PhD, have the wrong background (so, no education within the required field), or an incomplete application (for example, no scientific publications included). This results in a new, often shorter, list of possible candidates. If there are no qualified applicants left, the position can be announced again, maybe with an extended deadline. It could also be decided to postpone the recruitment till a later point in time.

In the next step the applicants are assessed by an expert committee on their scientific, educational and scientific leadership qualities. This expert committee consist of three members: a professor/assoc professor from outside of Norway, a professor/assoc professor from a Norwegian university other than NTNU and a professor/assoc professor from the institute. The committee writes a report in which each applicant is assessed as qualified or not qualified. The ones that are qualified are ranked, where the one on place 1 is seen as being best qualified for the job.

The applicants ranked are invited for a trial lecture and an interview. The committee that assesses the trial lectures and performs the interviews, consists normally of the leader of the institute, the leader of the group the new position belongs to, a student representative and a human resource staff member. They extend the report of the expert committee with the information they have gathered, and they can change the ranking of the candidates. This extended report is sent to the faculty, where the recruitment committee can approve it or not. If it is approved the Human Resource department will send a job offer to the candidate ranked as 1. If he or she refuses the offer, the next candidate gets an offer. When a candidate has accepted, a contract is signed and the process is finished.

Task 1 (25 points)

Model the recruitment process described in the case using BPMN. Explain your modelling decisions.



The model presented here is a straightforward way of modeling the case. For the grading of the answer we will look at three aspects:

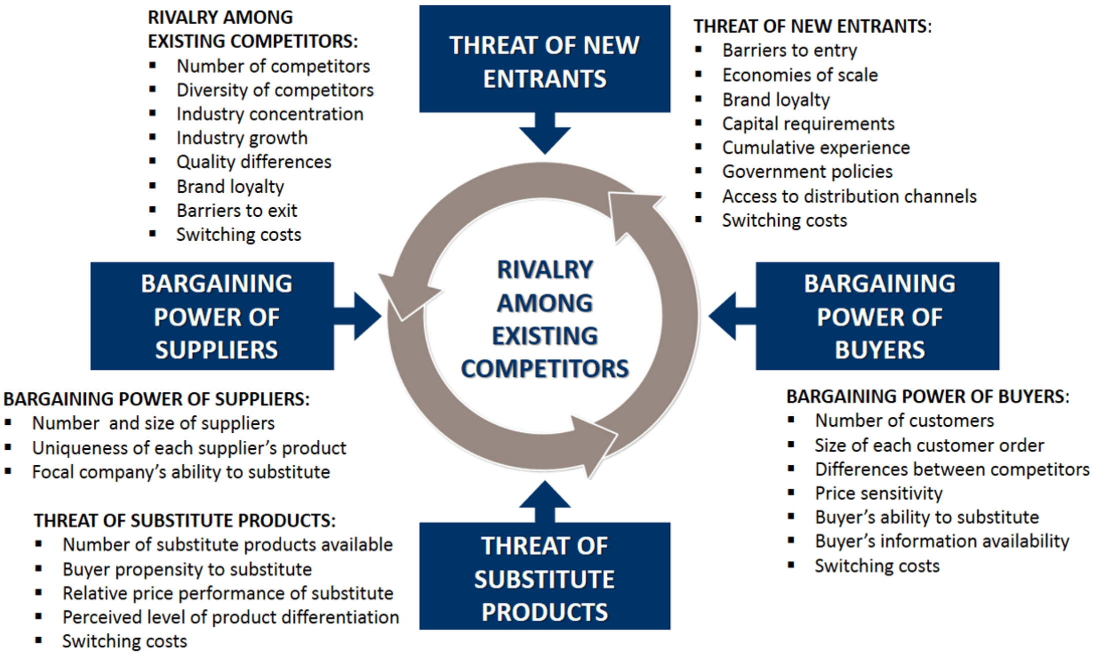
1. **Physical Quality** – this relates to *readability* of the model. Models with many control flows that go in all directions, or that have very many activities, or are difficult to read for other reasons, will get a reduction. In total will physical quality count for maximally 5 points.
2. **Syntactical Quality** – does the model adhere to the rules of BPMN? Are the modelling elements used correctly? An example of a violation of these rules is the appearance of so-called 'dangling' elements. All elements (except for data elements and data stores) must be on a control flow path connecting a start state to an end state. Syntactical quality counts for 10 points maximally.
3. **Semantical Quality** – is the model representing the textual description? This is of course often open for discussion, because the textual description only provides limited information on the process, and it is necessary to make assumptions and interpretations. However, logical consistency can be checked. Is the order of steps indicated correct (e.g in the example we have here, you can't have the expert committee begin assessing before the short list is made). Semantic Quality counts for maximally 10 points.

Task 2 (15 points)
Business analytical studies of the process have shown that there are possibilities for improvement. The average time from advertising the job announcement to signing an employer contract, is about 266 days, which is considered to be too long (**NB: these figures are made up and are not representative for NTNU)**. Main bottlenecks are around the work of the 'filtering' committee (takes on average: 24 days), and around the execution of trial lectures and interviews (average: 58 days). The reason why the filtering committee is a bottleneck is that it is difficult to find two people that want to do the job, and when having the job, they give it low priority. The reason why the trial lectures and interviews are a bottleneck is that the applicants have to come to NTNU to do these, and finding dates that fit is often problematic.

Suggest information systems that could help in removing/reducing these bottle necks. So, suggest the type of systems you would like to use, and how they would be used. Does implementing these systems change the process model from Task 1?

1. The first suggestion should concentrate on what is actually done by the filtering committee. The work of this committee offers possibilities for support of a decision support system or an expert system. (The last edition of the book uses the term expert system in stead of decision support system, as a more general concept). The student should note that the work to be done is quite structured (or at least semi-structured), which offers possibilities for automation. The second suggestion should concentrate on the need to meet physically, instead of facilitating a virtual meeting using some kind of collaborative software, such as Teams or Skype. Describing the right type of systems, that could help solving the mentioned problems gives 5 points.
2. For each of the two types of systems suggested, the student should describe how it would be used. So, for example, in the first case the student could say that the system marks candidates on the list as 'to-be removed' by checking the completeness of the application (are all required documents uploaded?) and their background (is their master in the right field?). An employee from the institute then checks these 'to-be removed' candidates and approves them by pressing a remove button. This produces the short list. So, it is important that (some of) the systems functionality and user interaction is specified. Maximum number of points to get here is also 5.
3. The third aspect of the answer should say something about changes to the process model. So, do activities disappear, change type, do actors disappear, etc. Also 5 points to win here.

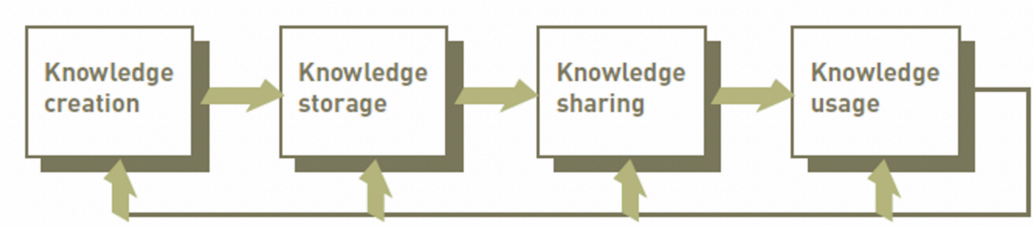
Task 3 (10 points)
One of the critical success factors for a university is that it succeeds in attracting the best scientists as their staff. Not being able to do so, affects its quality negatively. How would you position this threat in Porter's five forces model (see lecture Module 2.1)? If posting positions at Jobbnorge.no is the only strategy chosen by NTNU to guard itself against this threat, how would you assess this choice? Sufficient? Not sufficient? If not sufficient, which other strategies do you suggest? Motivate your answer.



1. If we look at Porters Five Forces Model you can say that it is mostly designed to analyze industrial markets, but it is definitely also applicable to the world of higher education and research. Even though all the strategies to cope with the powers, listed as bullets under each power, might be different here. The threat described in the task can be seen as an example of *rivalry among existing competitors* or *bargaining power of suppliers.* The first one is probably most obvious. Excellent academic researchers can work at many places. There is strong competition among universities to get them. *The bargaining power of suppliers* refers in our case to the possibility of high standing academics to negotiate high salaries and expensive research infrastructure in order to accept a position. 5 points for choice of threat/power and motivation.
2. The strategy chosen is *not* sufficient, I would say. More activities are needed to get broader attention for open positions. Students can suggest several additional advertisement strategies. Other ideas could be to raise salaries, to be more competitive, and invest in research infrastructure (labs, equipment). 5 points can be gained here.

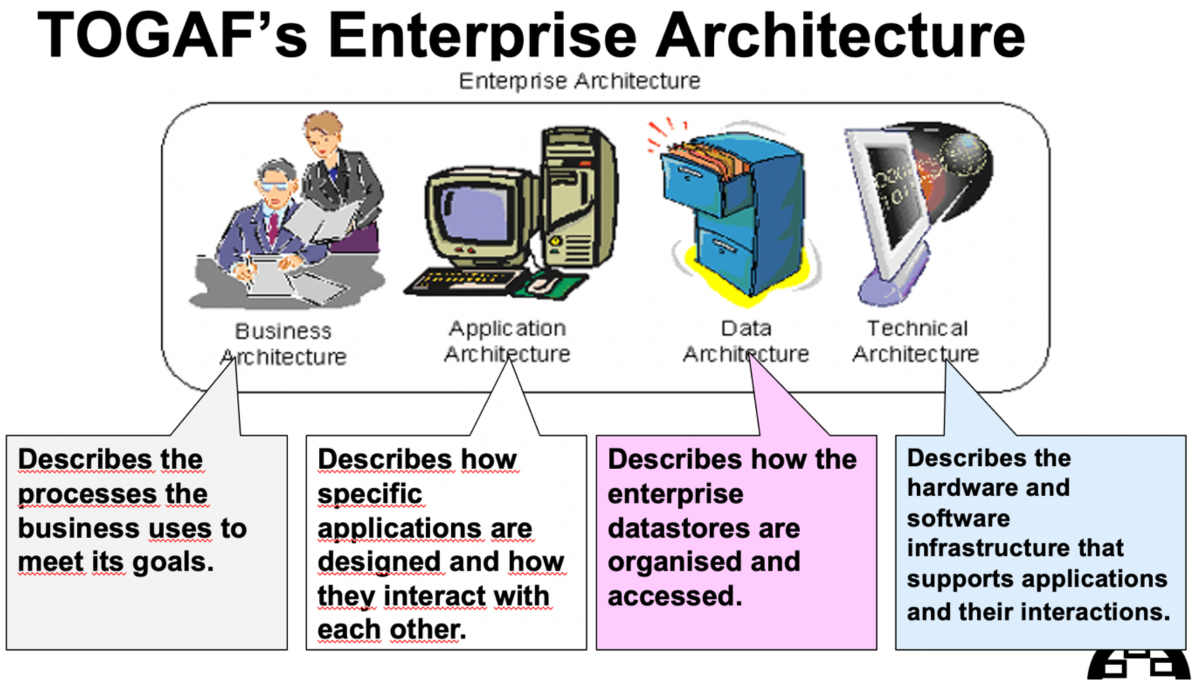
Task 4 (15 points)
Google scholar (scholar.google.com) is a useful tool to find scientific publications on a topic of interest. It can also be used to find publications of a specific person, as well as statistics about how many publications this person has and his or her so-called citation index. This citation index is often used as an indication of scientific quality. Could this system be characterized as a knowledge management system? And in which part of the recruitment process you modeled, can this system be used? Explain your answer.

- Knowledge management system (KMS):
  - ✓ Organized collection of people, procedures, software, databases, and devices
  - ✓ Used to create, store, share, and use the organization's knowledge and experience



1. The student should present a good analysis answering the question whether google scholar can be seen as a knowledge management system, or not. When they use the definition/description from the book, they can reflect on the fact that it is not for one organization but global. It supports knowledge sharing in communities of practice (scientists working in different fields). It focusses mostly on explicit knowledge storage and sharing. Tacit knowledge is not in focus. The student must show an understanding of what a knowledge management system is, and to what extend google scholar can be characterized as one. Can gain 10 points.
2. The most logical place in the process model where it can be used, is as a supportive system for the work of the expert cie, more specific the *evaluate short list candidates* activity. 5 points

Task 5 (15 points)
NTNU has decided to digitalize the recruitment process as much as possible. An Enterprise Architecture (EA), as discussed in the lecture by John Krogstie, can be a valuable tool in such an effort. Describe how an EA can be used in this case, and discuss at least two benefits of using it.



1. Based on feedback from students on the available learning material on EA we decided to give 8 of the 15 points for free.
2. The student should show an understanding what an EA is and what its purpose is. This could be done by using the four architectures used in TOGAF, but other ways are possible also. Main point must be that different models/descriptions are made of an organization's IS infrastructure, and that these models are used to control and align changes to this infrastructure. 5 points can be gained here.
3. Here are there many possibilities. Any reasonable, logical answer gets 2 points.